KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

YEAR 10 TRANSITION · 2026

Year 10 Mathematics Yearly Test

NAME _____ CLASS _____

TEACHER _____ DATE _____

WORKING TIME	TOTAL MARKS	CALCULATORS	BRING
1 hour 15 min	**55 marks**	**Allowed**	**Pen, pencil**
75 minutes	35 questions	bring your own	ruler, eraser, calculator

WHAT IS IN THIS TEST

	Section and topics	Questions	Marks	Time	Marker use
A	**Financial mathematics** Percentages, GST, profit, discount, wages, loans, interest	1–11	**15**	20 min	_____ / 15
B	**Measurement** Units, perimeter, circles, area, surface area, volume	12–21	**15**	20 min	_____ / 15
C	**Statistics and probability** Types of data, stem-and-leaf, mean, median, graphs, probability	22–29	**14**	20 min	_____ / 14
D	**Other Year 10 topics** Index laws, algebra, equations, straight-line graphs, Pythagoras, trigonometry	30–35	**11**	15 min	_____ / 11
	Total	**1–35**	**55**	**75 min**	_____ / 55

INSTRUCTIONS

1. Use a **blue or black pen.** Use a pencil and ruler for graphs and diagrams.
2. **Answer every question.** Write your answer on the **Answer** line, or in the table, box or graph.
3. **Show your working.** You can get marks for working, even if your answer is wrong.
4. **Multiple choice:** circle **one** letter, A, B, C or D.
5. Watch the clock. Each section has a **suggested time.**

STUCK? LOOK FOR HELP

Many questions have a **HELP** box. It gives you the **formula** and a worked **example**, like this:

HELP

Formula Speed = distance ÷ time

Example 100 km in 2 hours:
100 ÷ 2 = 50 km/h

Do not turn over until your teacher tells you to start. Good luck!

SECTION A

Financial mathematics

Questions **1–11** · **15 marks** · Suggested time **20 minutes**

____ / 15
Marker

Questions 1–4 are **multiple choice**. Circle the letter of the correct answer, like this: (B)

1 What is 28% of $89? **1 mark**

(A) $28.00 (B) $24.92 (C) $64.08 (D) $113.92

HELP **Example** 10% of $50: 0.10 × 50 = $5

2 A price tag says "**$55 (GST inc.)**". How much GST is included in the price? **1 mark**

(A) $5.00 (B) $5.50 (C) $50.00 (D) $60.50

HELP **Remember** GST in a price (GST inc.) = price ÷ 11

3 A shirt costs $56. It is **discounted by 26%**. What is the new price? **1 mark**

(A) $14.56 (B) $26.56 (C) $30.00 (D) $41.44

HELP **Remember** Find 26% of $56, then **take it off** the price.

4 Find the **simple interest** on $600 at 5% per year for 4 years. **1 mark**

(A) $12.00 (B) $30.00 (C) $120.00 (D) $720.00

HELP **Formula** $I = P \times R \times N$ (5% → 0.05)

5 Increase $968 by 12%. Use a multiplier and your calculator. **1 mark**

Working

HELP

Multiplier Increase by 8% → × 1.08
Decrease by 8% → × 0.92

Example Increase $50 by 10%: 50 × 1.10 = $55

Answer $ _____

Section A continues on the next page →

6 A shop buys a cap for \$7.60. This is the **cost price**. The shop adds a **50% mark-up**. **2 marks**

a Find the selling price of the cap.

Working

Answer \$ _____

b How much profit does the shop make?

Working

Answer \$ _____

HELP

Remember Mark-up of 30% → × 1.3
Profit = selling price − cost price

Example Cost \$10, mark-up 20%: 10 × 1.2 = \$12, profit = \$2

7 Tom is paid \$15.78 per hour. **2 marks**

a Tom works 30 hours. How much is he paid?

Working

Answer \$ _____

b Tom also works 4 hours of overtime at **time and a half**. How much is he paid for the overtime?

Working

Answer \$ _____

HELP

Remember Pay = hours × hourly rate
Time and a half = 1.5 × hourly rate

Example \$20 per hour, 3 hours at time and a half: 3 × 1.5 × 20 = \$90

8 Pedro sells phones. He earns **4.5% commission** on his sales. His sales this week are \$790. How much commission does he earn? **1 mark**

Working

Answer \$ _____

HELP

Remember 4.5% → 0.045
Commission = 0.045 × sales

9 Tranh works 36 hours at $28.89 per hour. **2 marks**

a Find Tranh's **gross pay** (total pay before anything is taken out).

Working

Answer $ _____

b Tranh pays $142.59 in tax and $22.50 in super (superannuation). Find his **net pay** (the pay he takes home).

Working

Answer $ _____

HELP

Remember Gross pay = hours × hourly rate
Net pay = gross pay − tax − super

10 Mia buys a phone on a payment plan. She borrows $1 200 and repays **$60 a month for 2 years**. How much **interest** does she pay? **1 mark**

Working

Answer $ _____

HELP

Remember 2 years = 24 months
Total repaid = repayment × number of months
Interest = total repaid − amount borrowed

Example Borrow $500, repay $50 a month for 1 year:
50 × 12 = $600
Interest = 600 − 500 = $100

11 $5 000 is invested for 3 years. It earns 4% **compound interest** each year. **2 marks**

a Find the value of the investment after 3 years.

Working

Answer $ _____

b How much interest is earned?

Working

Answer $ _____

HELP

Formula $A = P(1 + R)^n$
P = amount invested, R = rate as a decimal, n = number of years

Remember 5% → 1 + 0.05 = 1.05

Example $2 000 at 5% for 2 years:
$A = 2000 \times 1.05^2 = \$2\,205$
Interest = 2205 − 2000 = $205

Calculator Type 2000 × 1.05 ^ 2 =

SECTION B

Measurement

Questions **12–21** · **15 marks** · Suggested time **20 minutes**

_____ / 15
Marker

Questions 12–15 are **multiple choice**. Circle the letter of the correct answer, like this: (B)

12 How many metres are in 1 kilometre? **1 mark**

(A) 10 (B) 100 (C) 1 000 (D) 10 000

13 What is the **area** of a rectangle 3 cm long and 4 cm wide? **1 mark**

(A) 7 cm^2 (B) 12 cm (C) 14 cm (D) 12 cm^2

14 What is the **volume** of a cube with sides 3 cm long? **1 mark**

(A) 9 cm^3 (B) 27 cm^3 (C) 27 cm^2 (D) 54 cm^3

HELP **Formula** Volume of a cube = side × side × side

15 Which of these is 232 000 written in **scientific notation**? **1 mark**

(A) 2.32×10^5 (B) 23.2×10^4

(C) 2.32×10^{-5} (D) 232×10^3

HELP **Example** $2\,350\,000 = 2.35 \times 10^6$ (one digit before the decimal point)

16 Convert each measurement. **2 marks**

a Change 5.3 km into metres.

Answer _____ m

b Change 2.5 L into millilitres.

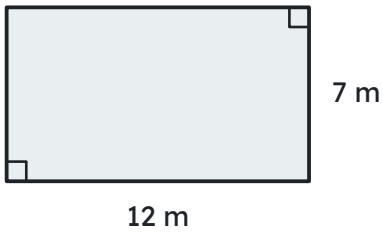
Answer _____ mL

HELP

Remember km → m: × 1000
L → mL: × 1000

17 A farmer puts a fence all the way around this rectangular paddock. How many metres of fence are needed? **1 mark**

HELP
Remember Perimeter = add up the lengths of **all** the sides.



Working

Answer _____ m

18 This circle has a radius of 3 m. Give your answers to 1 decimal place. **2 marks**

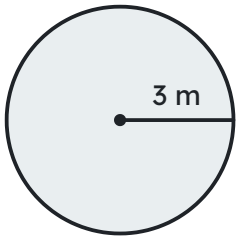
a Find the **circumference**.

Working

Answer _____ m

HELP
Formula $C = 2 \times \pi \times r$ (circumference)
$A = \pi \times r \times r$ (area)
Example Radius 5 cm:
$C = 2 \times \pi \times 5 = 31.4$ cm
$A = \pi \times 5 \times 5 = 78.5$ cm^2

b Find the **area**.

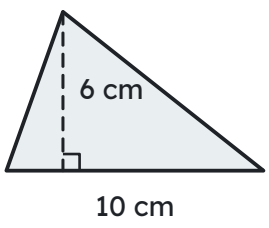


Working

Answer _____ m^2

19 Find the area of each shape. **2 marks**

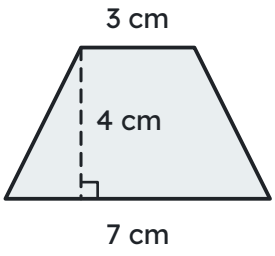
a Triangle



Working

Answer _____ cm^2

b Trapezium

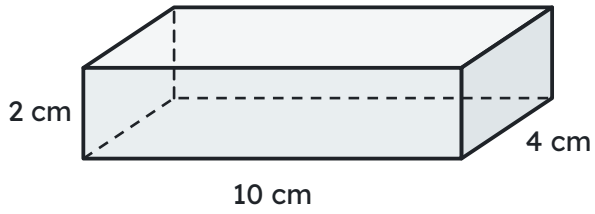


Working

Answer _____ cm^2

HELP
Formula Triangle: $A = ½ \times$ base $\times$ height
Trapezium: $A = ½ \times$ (top + bottom) $\times$ height
(top and bottom are the parallel sides)
Example Triangle, base 8 cm, height 5 cm:
$½ \times 8 \times 5 = 20$ cm^2

20 Find the **surface area** of this box. Complete the table. The first row is done for you. **2 marks**



HELP

Remember A box has 6 faces, in 3 matching pairs. Surface area = add the areas of all 6 faces.

Faces	Area of one face	Number of faces	Area
Top and bottom	$10 \times 4 = 40$ cm^2	2	80 cm^2
Front and back		2	
The two ends		2	
Surface area (add the last column)			

21 This cylinder has a radius of 2 cm and a height of 6 cm. **2 marks**

a Find the **volume**, to 1 decimal place.

Working

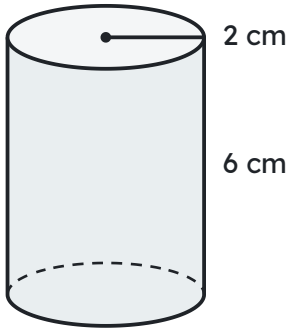
HELP

Formula Volume of a cylinder: $V = \pi \times r \times r \times h$

Example $r = 3$ cm, $h = 10$ cm: $V = \pi \times 3 \times 3 \times 10 = 282.7$ cm^3

Remember 1 cm^3 holds 1 mL

Answer _____ cm^3



b How many millilitres (mL) of water can it hold?

Working

Answer _____ mL

SECTION C

Statistics and probability

Questions **22–29** · **14 marks** · Suggested time **20 minutes**

____ / 14
Marker

Questions 22–25 are **multiple choice**. Circle the letter of the correct answer, like this: (B)

22 A survey asks: **"What is your favourite sport to play?"** What type of data does this give? **1 mark**

(A) numerical and discrete

(B) numerical and continuous

(C) categorical and ordinal

(D) categorical and nominal

HELP

Remember Categorical = words or groups. Numerical = numbers.
Nominal = no order (eye colour, type of pet).
Ordinal = has an order (small, medium, large).

Use this **stem-and-leaf plot** for Questions **23** and **24**.
It shows the scores of 11 players in a game.

Stem	Leaf
2	4 9
3	1 1 7 8
4	2 4 6
5	0 4

4 | 2 means 42

23 What is the **lowest** score? **1 mark**

(A) 4 (B) 20 (C) 24 (D) 54

24 What is the **mode** (the most common score)? **1 mark**

(A) 3 (B) 31 (C) 38 (D) 42

25 A normal 6-sided dice is rolled once. What is the probability of rolling an **even** number? **1 mark**

(A) $\frac{1}{6}$ (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) $\frac{2}{3}$

HELP **Formula** Probability = number of ways it can happen ÷ total number of ways

Section C continues on the next page →

26 These are the goals a netball player scored in 7 games: **3 marks**

8, 3, 13, 1, 5, 3, 9

Write the numbers in order first: ☐ ☐ ☐ ☐ ☐ ☐ ☐

a Find the **mean**.

Working

Answer _____

b Find the **median**. **Answer** _____

c Find the **range**. **Answer** _____

HELP

Mean	Add up the numbers, then divide by how many numbers there are
Median	The middle number, when they are in order
Range	Biggest – smallest

27 One letter is chosen at random from the word **2 marks**

I	N	T	E	R	E	S	T

a Find the probability that the letter is an **E**.

Answer _____

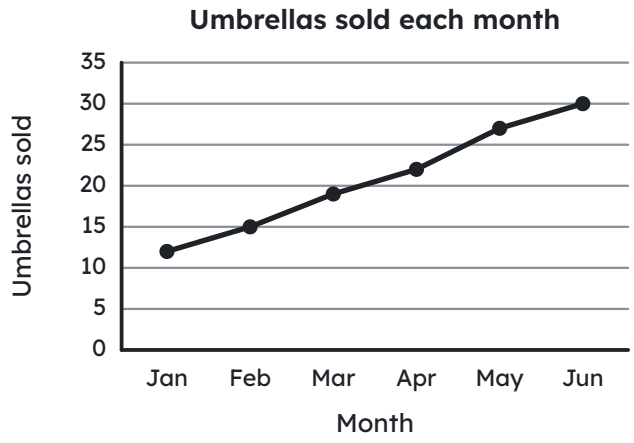
b Find the probability that the letter is a **vowel** (A, E, I, O or U).

Answer _____

HELP

Formula	Probability = number of ways it can happen ÷ total number of ways
Example	A letter from **BOOK**: $P(O) = \frac{2}{4} = \frac{1}{2}$

28 This **time-series graph** shows how many umbrellas a shop sold each month. **1 mark**



HELP

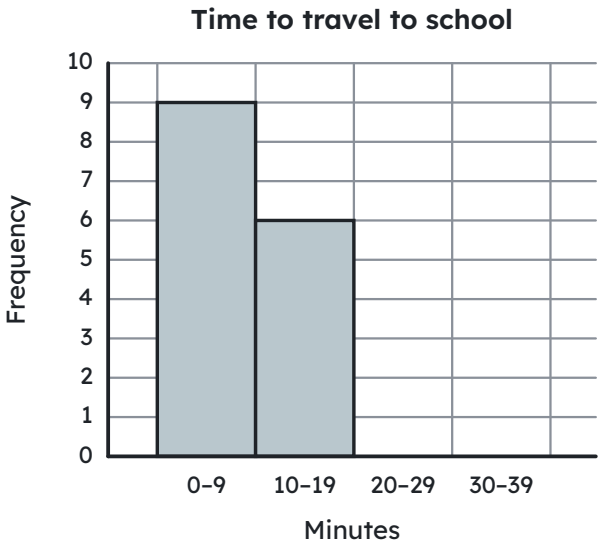
Remember	Upward trend: the line goes up over time. Downward trend: the line goes down over time.

Describe the **trend**. Circle one.

upward trend **downward trend** **no trend**

29 Some students recorded how many minutes it takes them to travel to school. **4 marks**

Minutes	Frequency
0–9	9
10–19	6
20–29	3
30–39	2
Total	



a Fill in the **total** at the bottom of the table.

b The graph has 2 columns drawn. Use the table to **draw the columns** for 20–29 and 30–39 on the graph.

c What **percentage** of the students took 0–9 minutes?

Working

HELP

Formula Percentage = frequency ÷ total × 100

Shapes



Answer _____ %

d Is the data **symmetrical** or **skewed**? Circle one.

symmetrical skewed

SECTION D

Other Year 10 topics

Questions **30–35** · **11 marks** · Suggested time **15 minutes**

_____ / 11
Marker

Question 30 is **multiple choice**. Circle the letter of the correct answer, like this: (B)

30 Simplify $a^3 \times a^4$ — **1 mark**

(A) a^7 (B) a^{12} (C) $2a^7$ (D) $7a$

HELP **Remember** When you multiply, **add** the powers. $m^2 \times m^6 = m^8$

31 If $a = 4$, find the value of $5a - 2$. — **1 mark**

Working

HELP
Remember $5a$ means $5 \times a$
Example If $a = 3$:
$2a + 1 = 2 \times 3 + 1 = 7$

Answer _____

32 Solve each equation. — **2 marks**

a $2x + 3 = 13$

Working

Answer $x =$ _____

HELP
Remember Do the **opposite** to both sides.
Example $3x + 1 = 13$
Take away 1: $3x = 12$
Divide by 3: $x = 4$
Example $\frac{x+1}{2} = 4$
Times by 2: $x + 1 = 8$
Take away 1: $x = 7$

b $\frac{x+2}{3} = 4$

Working

Answer $x =$ _____

Section D continues on the next page →

33 The rule for a straight line is $y = 2x + 1$. **3 marks**

a Use the rule to find y when $x = 0$, 1, 2 and 3. Write each answer in the table. 1 mark

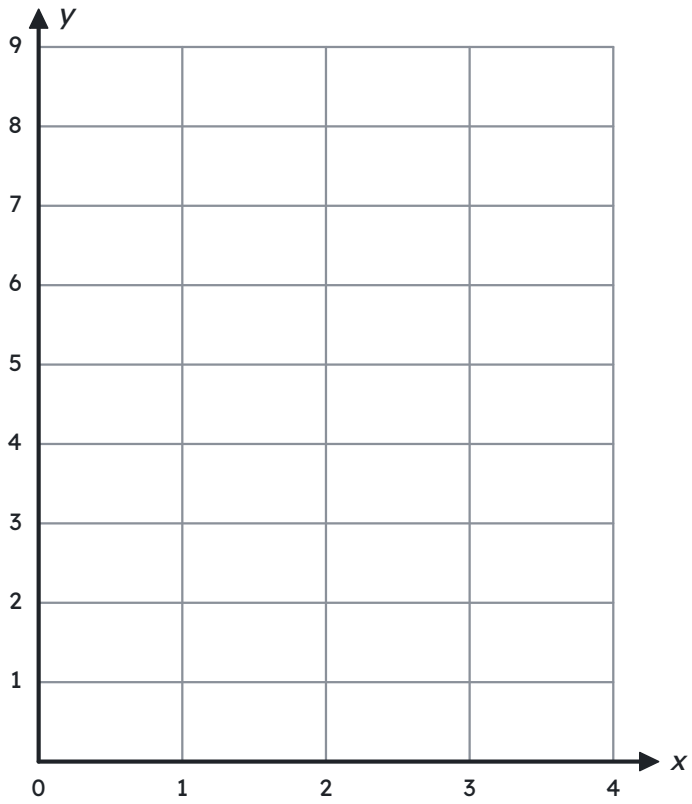
x	0	1	2	3
y				

b Plot the points on the grid and join them with a straight line. 1 mark

c Write the **gradient** and the ***y*-intercept** of $y = 2x + 1$. 1 mark

Gradient = _____

y-intercept = _____



HELP

Example For $y = 3x + 2$, when $x = 4$: $y = 3 \times 4 + 2 = 14$

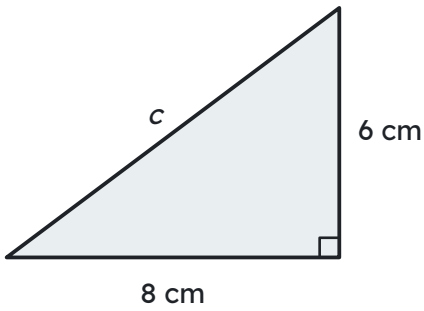
Plot For $x = 4$ and $y = 9$: go across to 4, then up to 9. Put a dot. Join the dots with a ruler.

$y = mx + c$ m is the gradient (how steep the line is). c is the y-intercept (where the line crosses the y-axis).

34 Find the length of the **hypotenuse**, *c*.

2 marks

Working



HELP

Formula $c^2 = a^2 + b^2$ (c is the longest side)

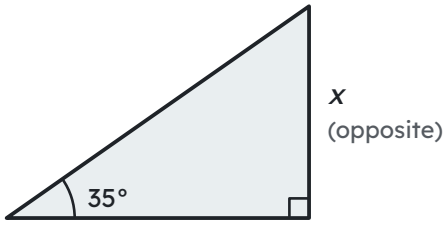
Example $a = 3$, $b = 4$:
$c^2 = 9 + 16 = 25$
$c = \sqrt{25} = 5$

Answer $c =$ _____ cm

35 Find the length of side *x*, to 1 decimal place.

2 marks

Working



HELP

Formula tan (angle) = opposite ÷ adjacent
so opposite = adjacent × tan (angle)

Example Angle 20°, adjacent 5 cm:
$x = 5 \times \tan 20° = 1.8$ cm

Calculator Check your calculator shows **D** or **DEG** (degrees).

Answer $x =$ _____ cm

End of the test

- Did you answer **every** question? Have a go at any you missed.
- Did you write the **units** ($, cm, m^2, mL) and show your **working**?
- Use the space below if you need more room. Write the question number next to your working.

Extra working space